

Лабораторная работа №13

Статическая маршрутизация в Интернете. Планирование

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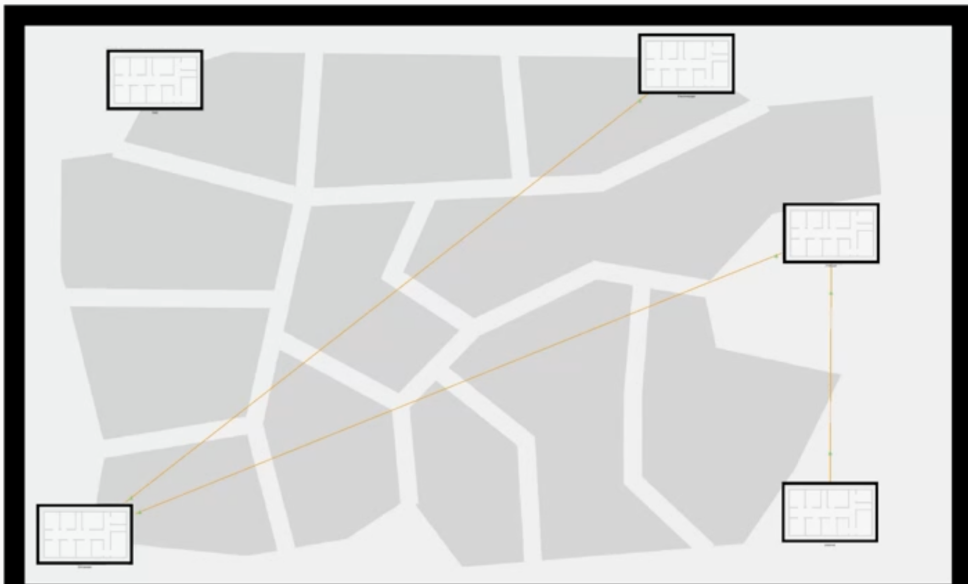
Информация

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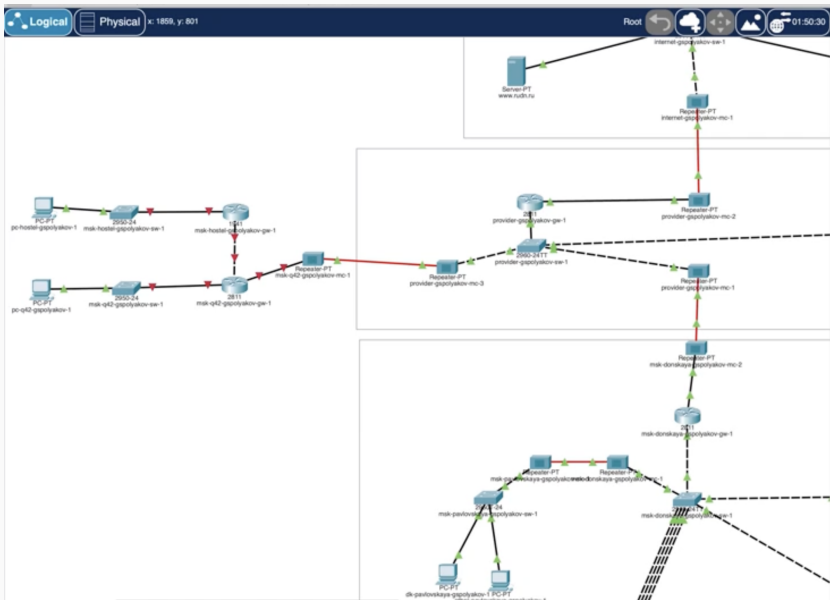
Выполнение работы







Маршрутизация




```
Router#conf
Router#configure ter
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#line vty 0 4
Router(config-line)#password cisco
Router(config-line)#login
Router(config-line)#exit
Router(config)#line console 0
Router(config-line)#password cisco
Router(config-line)#login
Router(config-line)#exit
Router(config)#secre
Router(config)#enabl
Router(config)#enable sect
Router(config)#enable se
Router(config)#enable secret cisco
Router(config)#servi
Router(config)#service pa
Router(config)#service password-encryption
Router(config)#user
Router(config)#username admin p
Router(config)#username admin pr
Router(config)#username admin privilege 1 se
Router(config)#username admin privilege 1 secret cisco
Router(config)#ip domain-name q42.rudn.edu
Router(config)#crypt
Router(config)#crypto key gene
Router(config)#crypto key generate rsa
% Please define a hostname other than Router.
Router(config)#hos
Router(config)#hostname nak-q42-gspolyakov-gw-1
nak-q42-gspolyakov-gw-1(config)#crypto key generate rsa
The name for the keys will be: nak-q42-gspolyakov-gw-1.q42.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

nak-q42-gspolyakov-gw-1(config)#line vty 0 4
*Mar 1 0:19:39.670: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:19:39.671: SSH-5-ENABLED: SSH 1.5 has been enabled
nak-q42-gspolyakov-gw-1(config-line)#tra
nak-q42-gspolyakov-gw-1(config-line)#transport in
nak-q42-gspolyakov-gw-1(config-line)#transport input rsa
%
% Invalid input detected at '' marker.
```

```
Switch>enable
Switch#conf
Switch#configure ter
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#host
Switch(config)#hostname mak-q42-gspolyakov-sw-1
mak-q42-gspolyakov-sw-1(config)# line vty 0 4
mak-q42-gspolyakov-sw-1(config-line)#password cisco
mak-q42-gspolyakov-sw-1(config-line)#login
mak-q42-gspolyakov-sw-1(config-line)#exit
mak-q42-gspolyakov-sw-1(config)#line console 0 4
^
% Invalid input detected at '' marker.

mak-q42-gspolyakov-sw-1(config)#line console 0
mak-q42-gspolyakov-sw-1(config-line)#password cisco
mak-q42-gspolyakov-sw-1(config-line)#login
mak-q42-gspolyakov-sw-1(config-line)#exit
mak-q42-gspolyakov-sw-1(config)#enable secret cisco
mak-q42-gspolyakov-sw-1(config)#service
mak-q42-gspolyakov-sw-1(config)#service pas
mak-q42-gspolyakov-sw-1(config)#service password-encryption
mak-q42-gspolyakov-sw-1(config)#username admin privele
mak-q42-gspolyakov-sw-1(config)#username admin priv
mak-q42-gspolyakov-sw-1(config)#username admin privilege 1 secret cisco
mak-q42-gspolyakov-sw-1(config)#ip domain-n
mak-q42-gspolyakov-sw-1(config)#ip domain-name q42.rudn.edu
mak-q42-gspolyakov-sw-1(config)#crypto k
mak-q42-gspolyakov-sw-1(config)#crypto key g
mak-q42-gspolyakov-sw-1(config)#crypto key generate r
mak-q42-gspolyakov-sw-1(config)#crypto key generate rsa
The name for the keys will be: mak-q42-gspolyakov-sw-1.q42.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

mak-q42-gspolyakov-sw-1(config)#line vty 0 4
*Mar 1 1:2:39.634: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 1:2:39.634: %SSH-5-ENABLED: SSH 1.5 has been enabled
mak-q42-gspolyakov-sw-1(config-line)#!r
mak-q42-gspolyakov-sw-1(config-line)#transport in
mak-q42-gspolyakov-sw-1(config-line)#transport input ssh
mak-q42-gspolyakov-sw-1(config-line)#!
```

```
Router>enable
Router#confi
Router#configure ter
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname
Router(config)#hostname mak-hostel-gspolyakov-gw-1
mak-hostel-gspolyakov-gw-1(config)#line vty 0 4
mak-hostel-gspolyakov-gw-1(config-line)#passw
mak-hostel-gspolyakov-gw-1(config-line)#password cisco
mak-hostel-gspolyakov-gw-1(config-line)#login
mak-hostel-gspolyakov-gw-1(config-line)#exit
mak-hostel-gspolyakov-gw-1(config)#line console 0
mak-hostel-gspolyakov-gw-1(config-line)#password cisco
mak-hostel-gspolyakov-gw-1(config-line)#login
mak-hostel-gspolyakov-gw-1(config-line)#exit
mak-hostel-gspolyakov-gw-1(config)#enable secret cisco
mak-hostel-gspolyakov-gw-1(config)#serv
mak-hostel-gspolyakov-gw-1(config)#service pa
mak-hostel-gspolyakov-gw-1(config)#service password-encryption
mak-hostel-gspolyakov-gw-1(config)#user
mak-hostel-gspolyakov-gw-1(config)#username admin pr
mak-hostel-gspolyakov-gw-1(config)#username admin privilege 1 secret cisco
mak-hostel-gspolyakov-gw-1(config)#ip ssh ver
mak-hostel-gspolyakov-gw-1(config)#ip ssh version 2
Please create RSA keys (of at least 768 bits size) to enable SSH v2.
mak-hostel-gspolyakov-gw-1(config)#ip doma
mak-hostel-gspolyakov-gw-1(config)#ip domain
mak-hostel-gspolyakov-gw-1(config)#ip domain-n
mak-hostel-gspolyakov-gw-1(config)#ip domain-name hostel.rudn.edu
mak-hostel-gspolyakov-gw-1(config)#crypto ke
mak-hostel-gspolyakov-gw-1(config)#crypto key ge
mak-hostel-gspolyakov-gw-1(config)#crypto key generate rsa
The name for the keys will be: mak-hostel-gspolyakov-gw-1.hostel.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

mak-hostel-gspolyakov-gw-1(config)#ip ssh version 2
*Mar 1 0:32:58.733: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:32:58.733: %SSH-5-ENABLED: SSH 1.5 has been enabled
Please create RSA keys (of at least 768 bits size) to enable SSH v2.
mak-hostel-gspolyakov-gw-1(config)#crypto key generate rsa
% You already have RSA keys defined named mak-hostel-gspolyakov-gw-1.hostel.rudn.edu ,
% Do you really want to replace them? [yes/no]: yes
```

```
VLINPROTO-1-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
VLINKE-3-UPDOWN: Interface FastEthernet0/2, changed state to down
VLINPROTO-1-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
VLINKE-5-CHANGED: Interface FastEthernet0/2, changed state to up
VLINPROTO-1-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Switch>ena
Switch>enable
Switch>conf
Switch>configure terminal
Switch>configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname
% Incomplete command.
Switch(config)#hostname mak-hostel-gspolyakov-sw-1
mak-hostel-gspolyakov-sw-1(config)#line vty 0 4
mak-hostel-gspolyakov-sw-1(config-line)#password cisco
mak-hostel-gspolyakov-sw-1(config-line)#login
mak-hostel-gspolyakov-sw-1(config-line)#exit
mak-hostel-gspolyakov-sw-1(config)#line console 0
mak-hostel-gspolyakov-sw-1(config-line)#pass
mak-hostel-gspolyakov-sw-1(config-line)#password cisco
mak-hostel-gspolyakov-sw-1(config-line)#login
mak-hostel-gspolyakov-sw-1(config-line)#exit
mak-hostel-gspolyakov-sw-1(config)#enable secret cisco
mak-hostel-gspolyakov-sw-1(config)#service pass
mak-hostel-gspolyakov-sw-1(config)#service password-encryption
mak-hostel-gspolyakov-sw-1(config)#user
mak-hostel-gspolyakov-sw-1(config)#username admin pri
mak-hostel-gspolyakov-sw-1(config)#username admin privilege 1 secret cisco
mak-hostel-gspolyakov-sw-1(config)#ip domain-na
mak-hostel-gspolyakov-sw-1(config)#ip domain-name hostel.rudn.edu
mak-hostel-gspolyakov-sw-1(config)#crypto key generate rsa
The name for the keys will be: mak-hostel-gspolyakov-sw-1.hostel.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

mak-hostel-gspolyakov-sw-1(config)#line vty 0 4
*Mar 1 11:43:36: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 11:43:36: %SSH-5-ENABLED: SSH 1.5 has been enabled
mak-hostel-gspolyakov-sw-1(config-line)#transport
```

```
VLINK-3-UPDOWN: Interface FastEthernet0/2, changed state to down

VLINKPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

VLINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

VLINKPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Switch>ena
Switch>enable
Switch#conf
Switch#configure ter
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname
% Incomplete command.
Switch(config)#hostname mak-hostel-gspolyakov-sw-1
mak-hostel-gspolyakov-sw-1(config)#line vty 0 4
mak-hostel-gspolyakov-sw-1(config-line)#password cisco
mak-hostel-gspolyakov-sw-1(config-line)#login
mak-hostel-gspolyakov-sw-1(config-line)#exit
mak-hostel-gspolyakov-sw-1(config)#line console 0
mak-hostel-gspolyakov-sw-1(config-line)#pass
mak-hostel-gspolyakov-sw-1(config-line)#password cisco
mak-hostel-gspolyakov-sw-1(config-line)#login
mak-hostel-gspolyakov-sw-1(config-line)#exit
mak-hostel-gspolyakov-sw-1(config)#enable secret cisco
mak-hostel-gspolyakov-sw-1(config)#service pass
mak-hostel-gspolyakov-sw-1(config)#service password-encryption
mak-hostel-gspolyakov-sw-1(config)#user
mak-hostel-gspolyakov-sw-1(config)#username admin pri
mak-hostel-gspolyakov-sw-1(config)#username admin privilege 1 secret cisco
mak-hostel-gspolyakov-sw-1(config)#ip domain-na
mak-hostel-gspolyakov-sw-1(config)#ip domain-name hostel.rudn.edu
mak-hostel-gspolyakov-sw-1(config)#crypto key generate rsa
The name for the keys will be: mak-hostel-gspolyakov-sw-1.hostel.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

mak-hostel-gspolyakov-sw-1(config)#line vty 0 4
*Mar 1 1:8:43.369: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 1:8:43.369: %SSH-5-ENABLED: SSH 1.5 has been enabled
mak-hostel-gspolyakov-sw-1(config-line)#transport in
mak-hostel-gspolyakov-sw-1(config-line)#transport input ssh
mak-hostel-gspolyakov-sw-1(config-line)#
```

```
Switch>enable
Switch#configure terminal
^
% Invalid input detected at '^' marker.

Switch#confi
Switch#configure t
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname
Switch(config)#hostname ech-sochi-gspolyakov-sw-1
ech-sochi-gspolyakov-sw-1(config)#line vty 0 4
ech-sochi-gspolyakov-sw-1(config-line)#password cisco
ech-sochi-gspolyakov-sw-1(config-line)#login
ech-sochi-gspolyakov-sw-1(config-line)#exit
ech-sochi-gspolyakov-sw-1(config)#line conso
ech-sochi-gspolyakov-sw-1(config)#line console 0
ech-sochi-gspolyakov-sw-1(config-line)#password cisco
ech-sochi-gspolyakov-sw-1(config-line)#login
ech-sochi-gspolyakov-sw-1(config-line)#exit
ech-sochi-gspolyakov-sw-1(config)#enable secr
ech-sochi-gspolyakov-sw-1(config)#enable secret cisco
ech-sochi-gspolyakov-sw-1(config)#servi
ech-sochi-gspolyakov-sw-1(config)#service pas
ech-sochi-gspolyakov-sw-1(config)#service password-encryption
ech-sochi-gspolyakov-sw-1(config)#user
ech-sochi-gspolyakov-sw-1(config)#username admin pr
ech-sochi-gspolyakov-sw-1(config)#username admin privilege 1 sec
ech-sochi-gspolyakov-sw-1(config)#username admin privilege 1 secret cisco
ech-sochi-gspolyakov-sw-1(config)#ip domain-n
ech-sochi-gspolyakov-sw-1(config)#ip domain-name sochi.rudn.edu
ech-sochi-gspolyakov-sw-1(config)#crypt
ech-sochi-gspolyakov-sw-1(config)#crypto key gene
ech-sochi-gspolyakov-sw-1(config)#crypto key generate rsa
The name for the keys will be: ech-sochi-gspolyakov-sw-1.sochi.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

ech-sochi-gspolyakov-sw-1(config)#line vty 0 4
*Mar 1 1:10:2.591: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 1:10:2.591: %SSH-5-ENABLED: SSH 1.5 has been enabled
ech-sochi-gspolyakov-sw-1(config-line)#tr
ech-sochi-gspolyakov-sw-1(config-line)#transport in
ech-sochi-gspolyakov-sw-1(config-line)#transport input ssh
ech-sochi-gspolyakov-sw-1(config-line)#
```

```
Router#conf t
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname sch-sochi-gspolyakov-gw-1
sch-sochi-gspolyakov-gw-1(config)#line vty 0 4
sch-sochi-gspolyakov-gw-1(config-line)#password cisco
sch-sochi-gspolyakov-gw-1(config-line)#login
sch-sochi-gspolyakov-gw-1(config-line)#exit
sch-sochi-gspolyakov-gw-1(config)#line con
sch-sochi-gspolyakov-gw-1(config)#line console 0
sch-sochi-gspolyakov-gw-1(config-line)#pass
sch-sochi-gspolyakov-gw-1(config-line)#password cisco
sch-sochi-gspolyakov-gw-1(config-line)#login
sch-sochi-gspolyakov-gw-1(config-line)#exit
sch-sochi-gspolyakov-gw-1(config)#asa
sch-sochi-gspolyakov-gw-1(config)#enable secr
sch-sochi-gspolyakov-gw-1(config)#enable secret cisco
sch-sochi-gspolyakov-gw-1(config)#servi
sch-sochi-gspolyakov-gw-1(config)#service pass
sch-sochi-gspolyakov-gw-1(config)#service password-encryption
sch-sochi-gspolyakov-gw-1(config)#user
sch-sochi-gspolyakov-gw-1(config)#username admin privele
sch-sochi-gspolyakov-gw-1(config)#username admin priv
sch-sochi-gspolyakov-gw-1(config)#username admin privilege 1 se
sch-sochi-gspolyakov-gw-1(config)#username admin privilege 1 secret cisco
sch-sochi-gspolyakov-gw-1(config)#ip domain-n
sch-sochi-gspolyakov-gw-1(config)#ip domain-name sochi.rudn.edu
sch-sochi-gspolyakov-gw-1(config)#cry
sch-sochi-gspolyakov-gw-1(config)#crypto k
sch-sochi-gspolyakov-gw-1(config)#crypto key g
sch-sochi-gspolyakov-gw-1(config)#crypto key generate r
sch-sochi-gspolyakov-gw-1(config)#crypto key generate rsa
The name for the keys will be: sch-sochi-gspolyakov-gw-1.sochi.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

sch-sochi-gspolyakov-gw-1(config)#lin
*Mar 1 1:12:43.597: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 1:12:43.597: SSH-5-ENABLED: SSH 1.5 has been enabled
sch-sochi-gspolyakov-gw-1(config)#line vty 0 4
sch-sochi-gspolyakov-gw-1(config-line)#tra
sch-sochi-gspolyakov-gw-1(config-line)#transport in
sch-sochi-gspolyakov-gw-1(config-line)#transport input as
sch-sochi-gspolyakov-gw-1(config-line)#transport input ssh
sch-sochi-gspolyakov-gw-1(config-line)#
```

